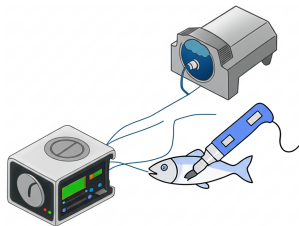


Deep Learning for Marine Biomass Analysis

Using Rapid Evaporative Ionization Mass Spectrometry



Jesse Wood

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Victoria University of Wellington

Supervisors: Bach Nguyen
Bing Xue
Mengjie Zhang
Daniel Killeen

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Figure 1. Hoki (Blue Grenadier)



Figure 2. Mackerel

The Big Picture: Why This Matters?

The Challenges:
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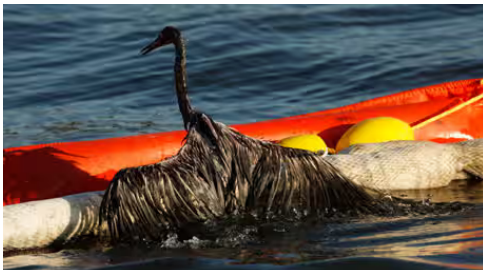
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Nutrition Facts	
6 servings per container	
Serving size 4-5 ounces(187g)	
Amount per serving	200
Calories	% Daily Value*
Total Fat 5g	6%
Saturated Fat 0.5g	3%
Trans Fat 0g	
Cholesterol 80mg	27%
Sodium 610mg	27%
Total Carbohydrate 10g	4%
Dietary Fiber 0g	0%
Total Sugars 3g	
Includes 0g Added Sugars	0%
Protein 27g	
Vitamin D 2mcg	10%
Calcium 79mg	6%
Iron 3mg	15%
Potassium 519mg	10%
*The % Daily Value tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.	



The Core Challenge: A Lack of Trust and Safety

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The screenshot shows the top portion of a news article on the Daily Mail Australia website. The page has a blue header with the 'Daily Mail AUSTRALIA' logo and a navigation bar with links to Home, U.K., U.S., News, World News, Sport, TV&Showbiz, Femail AU, Health, and Science. Below the navigation bar is a secondary bar with links to Monkeypox, Covid-19, Russia-Ukraine War, Breaking News, Sydney, Melbourne, Brisbane, and New Zealand. The main headline is 'Popular restaurant accused of serving cheap Vietnamese catfish to customers who thought they were getting Australian dory'. Below the headline is a bulleted list of key points. The article is attributed to HARRY PEARL FOR DAILY MAIL AUSTRALIA, with a publication date of 14:31 AEDT, 27 May 2016, and an update date of 16:08 AEDT, 27 May 2016. At the bottom of the article preview, there are social media sharing icons for Facebook, WhatsApp, Twitter, and Email, along with a 'Share' button, a '47 shares' counter, and a 'View comments' link with a comment icon and the number 9.

Daily Mail AUSTRALIA

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Popular restaurant accused of serving cheap Vietnamese catfish to customers who thought they were getting Australian dory

- A Melbourne restaurant has been accused of serving catfish to customers
- Hunky Dory has allegedly been selling frozen fillets of basa as dory
- Owner Greg Robotis has denied allegations he is misleading customers
- The City of Port Phillip is investigating Hunky Dory's Port Melbourne store

By HARRY PEARL FOR DAILY MAIL AUSTRALIA
PUBLISHED: 14:31 AEDT, 27 May 2016 | UPDATED: 16:08 AEDT, 27 May 2016

Share 47 shares View comments

A Melbourne restaurant has been accused of serving a Vietnamese catfish to customers who believe they are ordering Dory.

A whistleblower has alleged that Hunky Dory outlets have been selling frozen fillets of basa, a species of catfish native to the Mekong basin, as fish-of-the-day dory, **The Age** reports.

Owner Greg Robotis has denied the claims and said inexperienced staff may have been calling the fish the wrong name.

The Technical Hurdle: Identifying Samples

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A "Chemical Fingerprint": The Promise of REIMS

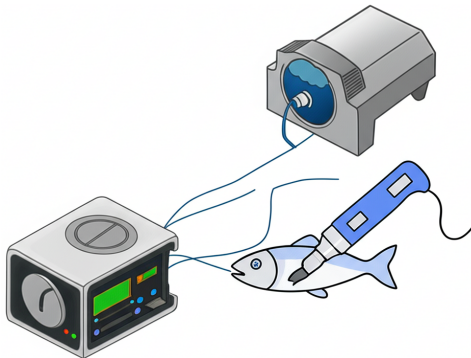
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The Data Problem: Complexity & Noise

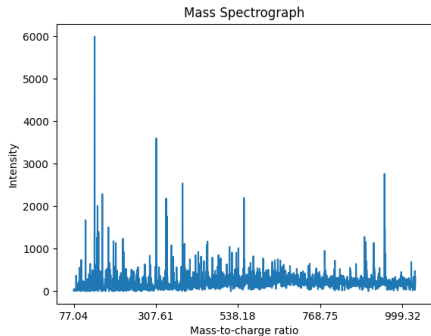
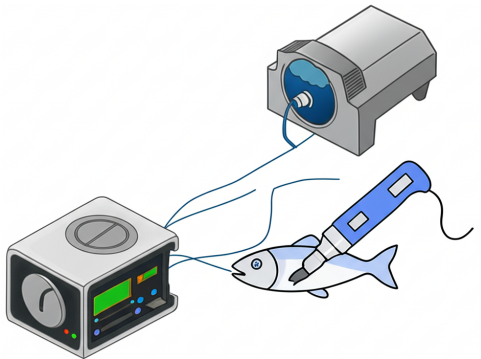
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My Solution: A "Fingerprint Reader"

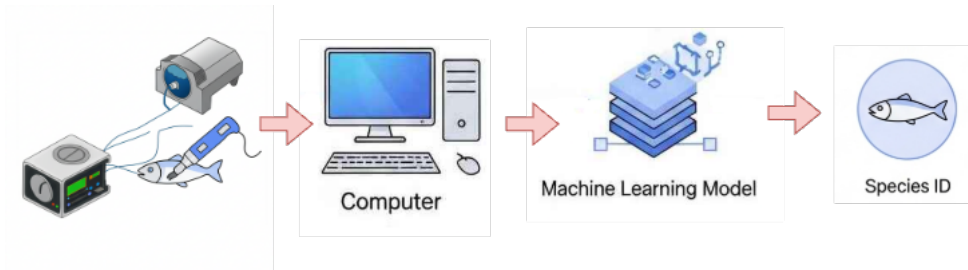
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The Right Tool for the Job: The Transformer

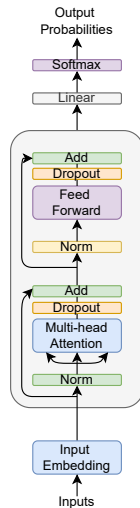
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How a Transformer "Reads" Spectra

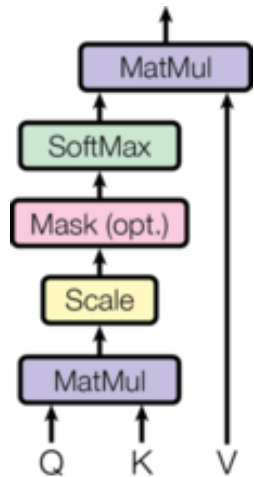
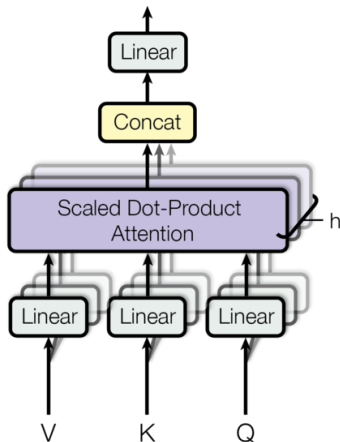
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Novel Method 1: Pre-training Strategy

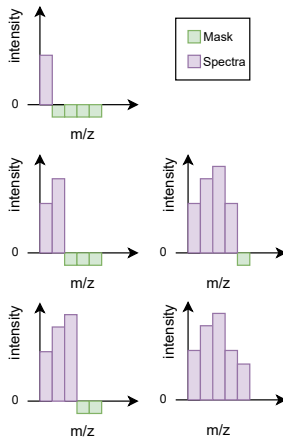
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Novel Method 2: Mixture of Experts Transformer

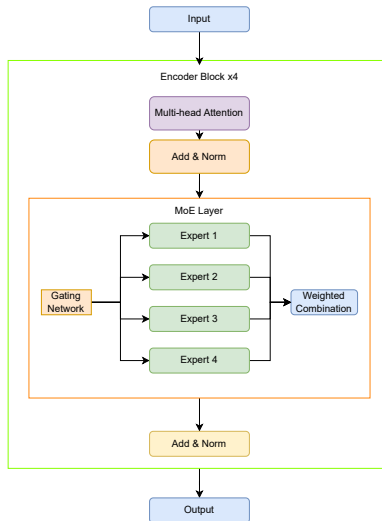
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Novel Method 3: Transfer Learning for Low-Data Tasks

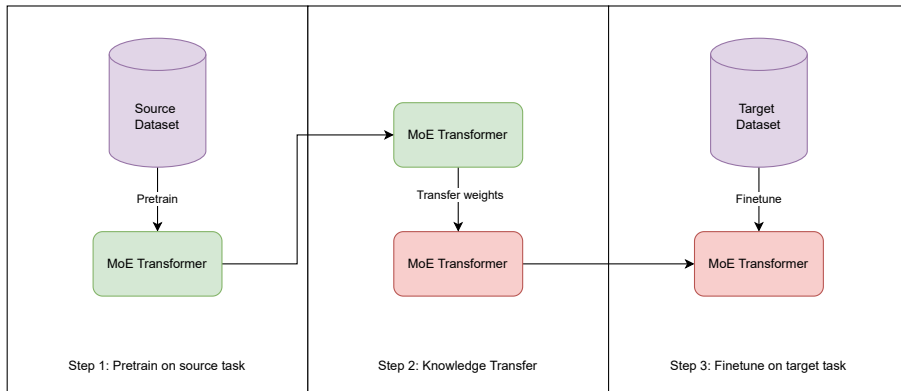
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Application 1: Fish Species Identification

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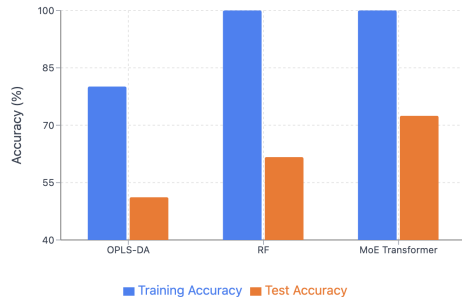
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Fish Body Parts Classification Accuracy Comparison



Application 2: Fish Body Part Identification

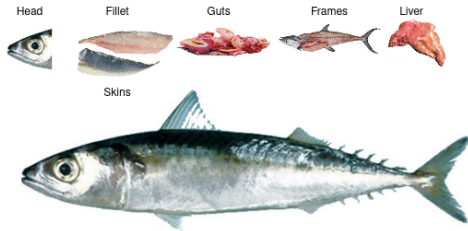
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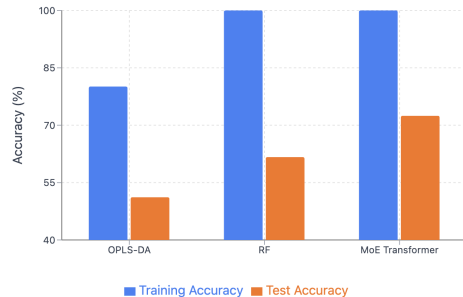
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Fish Body Parts Classification Accuracy Comparison



Application 3: Detecting Cross-Species Adulteration

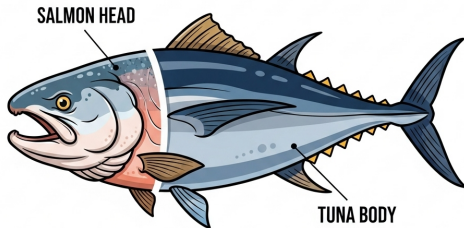
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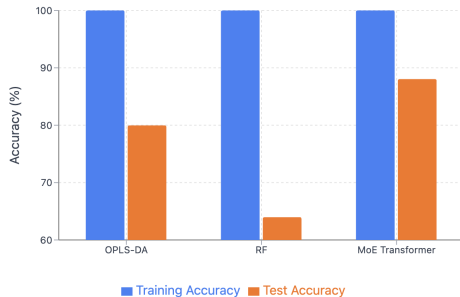
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Cross-species Adulteration Accuracy Comparison



Application 4: Detecting Oil Contamination

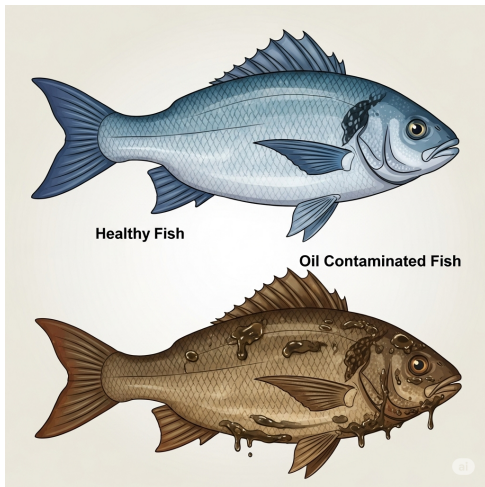
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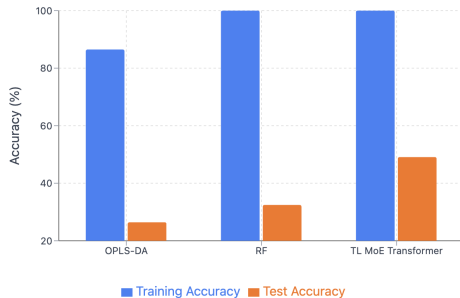
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Oil Contamination Detection Accuracy Comparison



Interim Summary: The "Fingerprint Reader" is a Success

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- ☒ Species ID (100%)
- ☒ Body Part ID (72%)
- ☒ Cross-species Adulteration (88%)
- ☒ Oil Contamination (48%)

A New Frontier: Identifying the Processing Batch

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The Task: A Pair-Wise Problem

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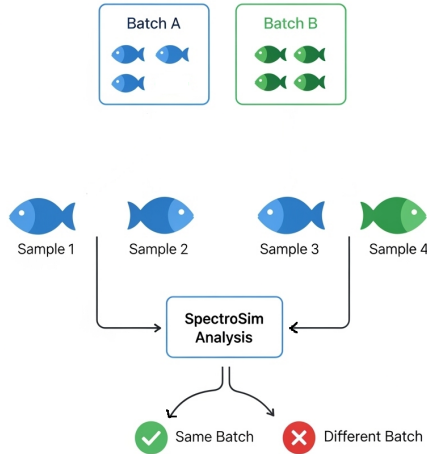
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The Old Way: Binary Classification

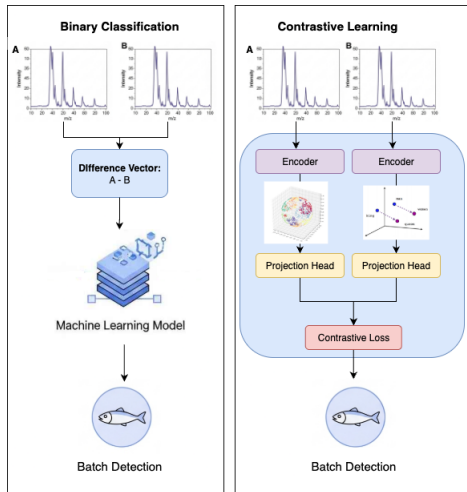
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My Solution: SpectroSim

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A self-supervised model to solve this without labels.

The SpectroSim Approach: Contrastive Learning

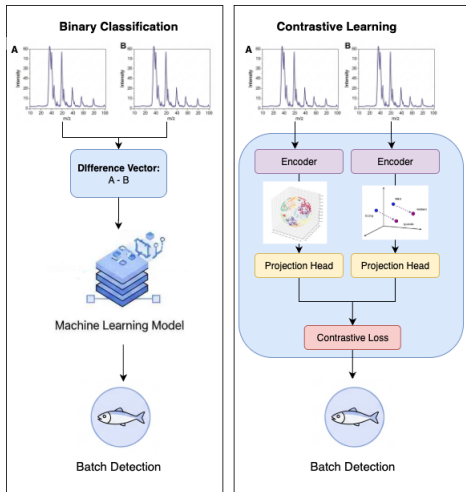
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Why Contrastive Learning is Better

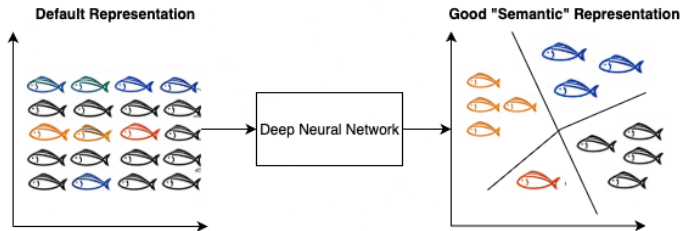
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The Old Architecture of SimCLR

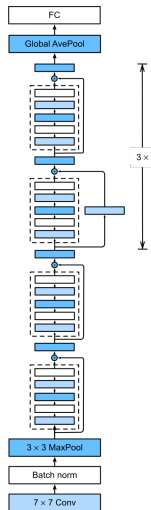
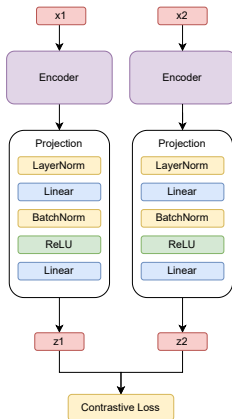
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The New Architecture of SpectroSim

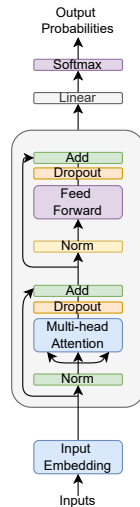
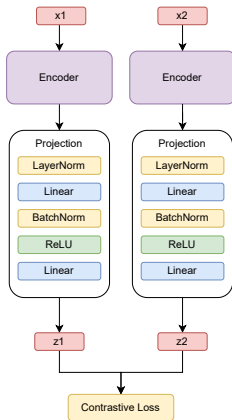
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SpectroSim: The Results

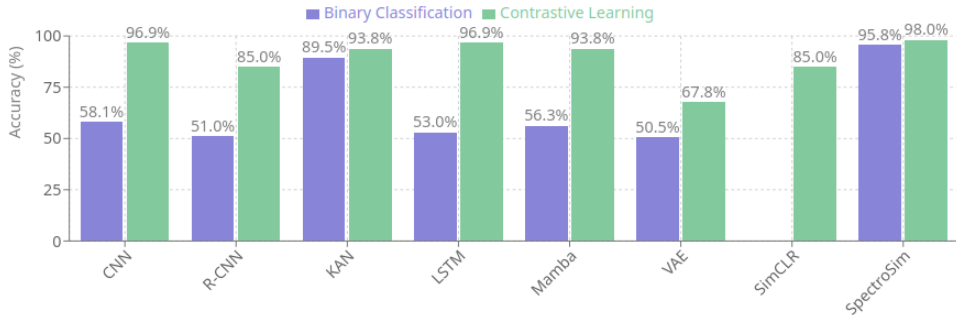
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SpectroSim vs The Rest

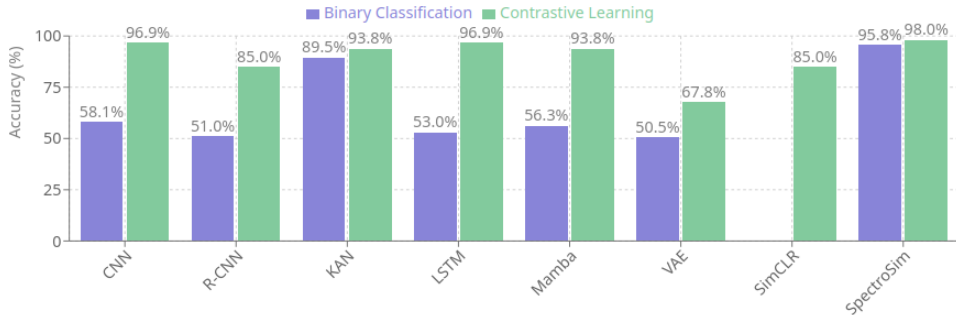
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Summary of Contributions: A Better "Fingerprint Reader"

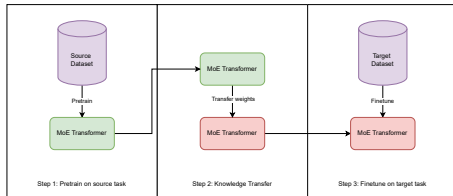
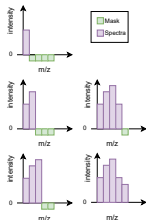
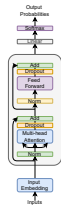
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Summary of Contributions: Solving the Unsolvable

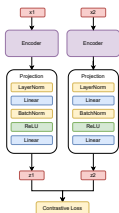
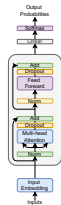
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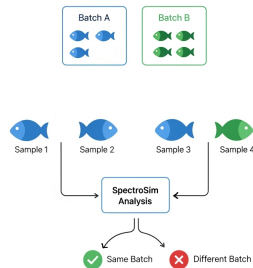
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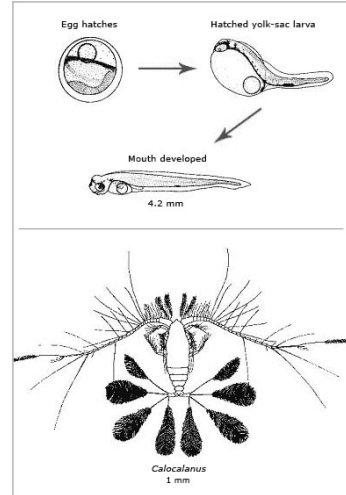
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Future Work

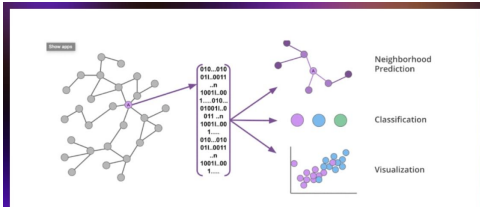
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Summary

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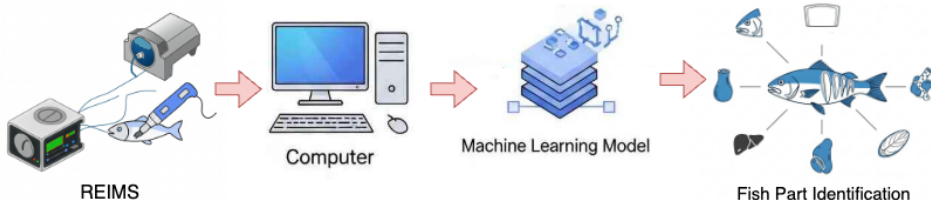
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I developed a suite of novel deep learning models that can rapidly and accurately decode chemical fingerprints from fish, solving critical challenges in food safety, fraud and quality control.



Thanks for your attention!

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